

Supratik Bhowal

+91 62895 52885

supratikbhowal23@gmail.com

[Google Scholar](#)

[LinkedIn](#)

[GitHub](#)

[Portfolio](#)

Research Interests

Mechanistic Interpretability, Computer Vision, AI Alignment, AI Safety, Scientific Machine Learning

Education

Institute of Engineering & Management (IEM), Kolkata

2024 – Present

B.Tech in Computer Science (AI & ML)

YGPA: 9.11 (2nd yr)

Experience

BlueDot Impact

Sep 2026 – Present

Technical AI Safety Fellow

Remote

Selected as a Technical AI Safety Fellow to build applied expertise in AI alignment, evaluations, mechanistic interpretability, AI control, and scalable oversight. Attend structured classes and role-play discussions with peer fellows to deepen understanding of alignment and oversight problems.

IEM GenAI Centre of Excellence

Aug 2026 – Present

Research Co-Lead

Kolkata, India

Leading two research groups under the society, setting project direction and reviewing research output across teams.

- Lead one group's project investigating why linear probes and sparse-autoencoder (SAE) features return contradictory verdicts on a quantized Gemma-2-2B, proposing a geometric signal-to-perturbation account to reconcile the disagreement [P6].
- Lead a second group building a null-model audit framework to test whether learned circuits (not just representations) survive quantization and weight pruning, benchmarking observed circuit drift against a frozen, matched-magnitude noise distribution and other rubrics [P7].

Rouge International & Rouge Ventures

May 2026 – Present

Associate Intern

Hong Kong (Remote)

Built and deployed AI automation tools for different functions across the VC firm, like research automation & lead sourcing. Led internal training sessions on using AI for better workflows, and helped teams build their own lightweight tools to solve function-specific problems.

Asvix

Apr 2026 – Present

AI Development Intern ([Offer Letter](#))

Remote

Engineered *DigiLab*, a domain-specific Media LLM chatbot for SOJNMS & IGNOU.

- Built the RAG, PDF-parsing, and knowledge-retrieval components, and tuned LLM response generation to improve retrieval accuracy on domain-specific queries.
- Worked with vector and graph databases (Pinecone, Neo4j) for RAG storage and retrieval, and tuned indexing/retrieval strategies for better contextual accuracy.

IEEE Systems, Man, and Cybernetics Society (SMCS), KGEC

May 2026 – Aug 2026

Summer Research Intern ([Certificate](#))

Kalyani, India

Supervisor: *Prof. Animesh Hazra*

- Adapted a frozen Phikon-v2 pathology foundation model with DoRA and self-supervised boundary signals for parameter-efficient histology instance segmentation (*Boundary-Aware JEPA*). Evaluated the approach across the GlaS, PanNuke, and CoNSeP histopathology datasets [P4].

Media Nirvana

Jan 2026 – June 2026

AI Intern ([Certificate](#)) ([LOR](#))

Hyderabad, India

Built end-to-end AI infrastructure for startup clients.

- Redesigned client websites, made custom CRM & WhatsApp bots, lead-capture systems and email automation for outreach.
- Developed *BrandSCOPE*, a multi-engine content engine that mines brand voice, tracks live trends via free APIs, and generates SEO/AEO/GEO-optimized articles end-to-end using local LLMs (Ollama) [7].

IEM with IIT Mandi Chapter

July 2025 – June 2026

HCI Research Intern

Kolkata, India

Supervisor: *Dr. Mousumi Laha*

- Conducted research in Human-Computer Interaction, focusing on innovative interfaces and robot training.
- Built ML and DL models for real-time brain EEG signal processing, using the models for real-time prediction [P5].

Supervisor: *Dr. Subhadip Chandra*

- Developed a complete autonomous driving stack for Indian traffic conditions with an eight-state FSM behaviour planner, LiDAR–Radar sensor fusion, and PID/Pure Pursuit control.

- Implemented an A* global router with smooth B-spline lane-change trajectory planning, achieving 77.8% route completion ($4.4\times$ CARLA’s BehaviourAgent) across a 30-route, 3-town benchmark [P1].

IEEE Engineering Medicine and Biology Society (EMBS)

Jun 2025 – Jul 2025

Student Research Intern (*Certificate*)

Pune, India

Supervisor: *Prof. Sneha Deshmukh*

- Developed and benchmarked CNN-BiLSTM and Transformer architectures for real-time epileptic seizure detection and prediction from EEG signals.

Selected Independent Research

- First-authored *Position, Not Provenance*, investigating whether chain-of-thought reasoning in medical vision-language models (LLaVA-Med, MedGemma) causally drives their answers or is merely decorative, using a perturbation-based causal-mediation pipeline on VQA-RAD; under review at ICVGIP 2026 [P2].

- Co-authored *Reliability Without Validity: Diagnosing Instrument Failure in LLM Preference Elicitation* at Apart Research’s Digital Minds Research Sprint, testing whether LLM “preferences” are genuine or artifacts of the elicitation method; compared forced-choice, reasoned-choice, and rating-scale protocols across two GPT-OSS models, finding instrument reliability to be model-dependent and refusals to mask different safety-relevant items per model. [Project page]

Publications**Published / Accepted**

- P1. Gope, S., **Bhowal, S.**, Gangopadhyay, A., Roy, A., Bakshi, A., Chandra, S. “A Formally Verified Autonomous Driving System for Indian Right-Hand Driving in CARLA” – Accepted at IPTA (IEEE International Conference on Image Processing Theory, Tools and Applications), 2026.

Under Review

- P2. **Bhowal, S.**, Basu, S., Mahanta, A. G., Chowdhury, A. P. “Position, Not Provenance: Separating Reasoning Mediation from Sycophancy in Medical Vision-Language Models” – Under Review at ICVGIP 2026. [arXiv:2607.27304](https://arxiv.org/abs/2607.27304).
- P3. **Bhowal, S.**, Gope, S., Mallik, B. B., Majumdar, S., Sengar, S. S. “AI Governance in Higher Education: Ethics, Policy, Assessment Integrity, and Human Capability – A Critical Literature Review” – Under review at Artificial Intelligence Review (Springer), 2026.

Manuscripts in Preparation

- P4. **Bhowal, S.** “Boundary-Aware JEPA: Parameter-Efficient Histology Instance Segmentation with Frozen Pathology Foundation Models” – Manuscript in preparation, 2026.
- P5. Gangopadhyay, A., **Bhowal, S.**, Gope, S. “Frequency-Aware Multi-Scale Deep Learning with Conditional WGAN-GP Augmentation for EEG Motor-Intent Decoding” – Manuscript in preparation, 2026.
- P6. **Bhowal, S.**, Roy, A., Kumari, I. “Reconciling Interpretability Instrument Disagreement Under Quantization” – Manuscript in preparation, 2026.
- P7. **Bhowal, S.**, Gope, S. “Do Circuits Survive Compression? A Null-Model Audit of Circuit Stability Under Quantization and Pruning” – Manuscript in preparation, 2026.

Skills

Programming: Python, MATLAB, Java, HTML5, Objective-C	ML/DL: PyTorch, TensorFlow, Scikit-learn, HuggingFace Transformers, OpenCV, Weights & Biases, JAX
Frameworks: LangChain, LangGraph, FastAPI, Jev	Tools: Git/GitHub, L ^A T _E X, Jupyter, GCP, MySQL

Projects**Agentic HR Operations Assistant** | *Python, NLP, RAG, Faiss, PyTorch* | [GitHub](#)

NLP-powered system that answers complex HR policy queries using RAG and executes HR actions from natural language.

- Combines a Faiss-indexed dense retrieval layer over HR policy documents with an intent-parsing layer that maps free-text requests into structured, executable HR actions.

HarmonyRL | *Python, HuggingFace, PyTorch, Scipy, Librosa, RL* | [GitHub](#)

Deep learning framework for symbolic music (MIDI) generation using supervised learning, reinforcement learning, and diffusion-based postprocessing.

- Pairs a supervised pretraining stage for note-sequence generation with an RL fine-tuning stage using custom harmonic/rhythmic reward shaping, followed by diffusion-based postprocessing for expressive humanization.

BrandSCOPE | *Python, LangChain, JavaScript, OpenAI API* | [GitHub](#)

Automated content engine that extracts a brand’s DNA, tracks live industry and brand news, builds an SEO/AEO/GEO-structured brief, writes the article in the brand’s voice, and delivers it – on a daily schedule with no human in the loop.